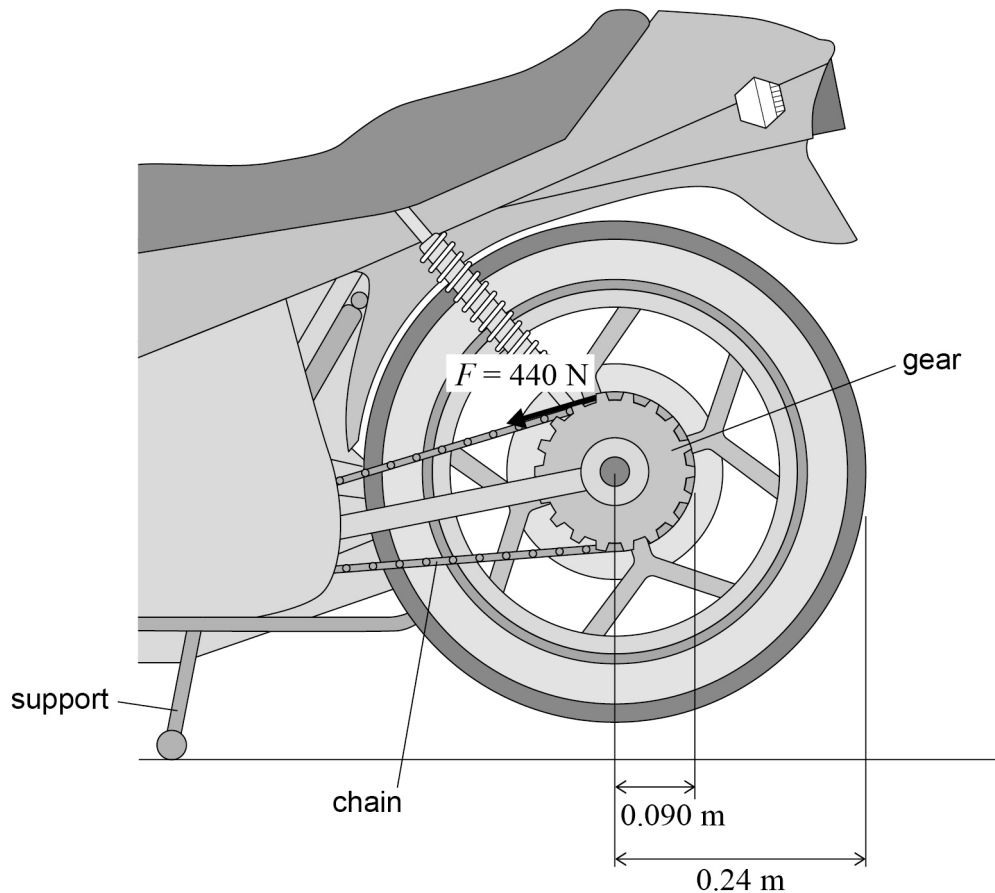


Figure 3 shows the rear wheel of a motorcycle. The motorcycle is supported so that the wheel does not touch the ground. The chain pulls on a gear to rotate the wheel.

Figure 3



The radius of the gear is 0.090 m and the force F exerted by the chain is 440 N. The wheel has a mass of 12 kg, a moment of inertia of 0.69 kg m^2 and a radius of 0.24 m.

The wheel experiences a resistive torque of 4.0 N m.

Calculate the angular acceleration of the wheel.

[3 marks]

angular acceleration = _____ rad s^{-2}

Question 3 continues on the next page

Turn over ►



0 3 . 2 Calculate the angular speed of the wheel after 0.60 s.

[1 mark]

angular speed = _____ rad s^{-1}

0 3 . 3 The motorcycle is taken off its support so that both wheels are on the ground. The motorcycle accelerates from rest. The wheels do not slip.

Show that the angular speed of the wheel is approximately 120 rad s^{-1} when the motorcycle is travelling at 28 m s^{-1} .

[1 mark]



0 3 . 4

The total mass of the motorcycle and rider is 290 kg. The front wheel has the same mass, dimensions and moment of inertia as the rear wheel.
The motorcycle travels at 28 m s^{-1} .

Determine the rotational kinetic energy of the wheels as a percentage of the total kinetic energy of the motorcycle.

[4 marks]

percentage = _____

0 3 . 5

An engineer suggests that the wheels should be replaced with wheels that have the same dimensions but are made from a material that is less dense.

Discuss how the change would affect the ability of the motorcycle to accelerate and decelerate.

[3 marks]

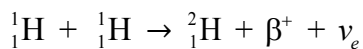


0 4 . 1 In an experimental nuclear fusion reactor, fusion happens in a plasma.

Explain how the properties of the plasma allow fusion reactions to happen.

[3 marks]

0 4 . 2 Two hydrogen nuclei fuse to produce deuterium as shown.



Determine, in J, the amount of energy released in this reaction.

mass of ${}^1_1\text{H} = 1.6726 \times 10^{-27} \text{ kg}$

mass of ${}^2_1\text{H} = 3.3435 \times 10^{-27} \text{ kg}$

mass of $\beta^+ = 0.0009 \times 10^{-27} \text{ kg}$

mass of ν_e is negligible

[3 marks]

energy released = _____ J



0 4 . 3

The reaction described in question **04.2** is the first part of the solar fusion (hydrogen) cycle.

Outline the other **two** reactions that occur in the hydrogen cycle.

[2 marks]

First reaction

Second reaction

8

Turn over for the next question

Turn over ►



0 5 . 1

A solar panel with an area of 1.0 m^2 is placed at the Earth's surface at an angle of 90° to the direction of the Sun's radiation. The output power of the Sun is $3.9 \times 10^{26} \text{ W}$. The radius of the Earth's orbit around the Sun is $1.5 \times 10^{11} \text{ m}$. The panel has an efficiency of 15%.

Show that the power output from the solar panel is approximately 200 W.

[2 marks]

0 5 . 2

A nuclear fission power station has an output power of 1.2 GW.

Calculate the number of solar panels, identical to the one in question **05.1**, required to produce the same output as the nuclear power station.

[1 mark]

number of panels = _____



0 5 . 3

State how the power output of an array of solar panels can be maximised throughout the day.

[1 mark]

0 5 . 4

Compare the environmental effects of the use of solar panels with the environmental effects of a nuclear fission power station.

[3 marks]

7

END OF SECTION A

Turn over ►



Section B

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.



If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional sheets for this working.

3**2**

A bicycle wheel has a rotational kinetic energy E when the bicycle is travelling at 8.0 m s^{-1} . The speed of the bicycle changes to 12.0 m s^{-1} .

What is the new rotational kinetic energy of the wheel?

[1 mark]

A $0.44E$

☐

B $1.22E$

☐

C $1.50E$

☐

D $2.25E$

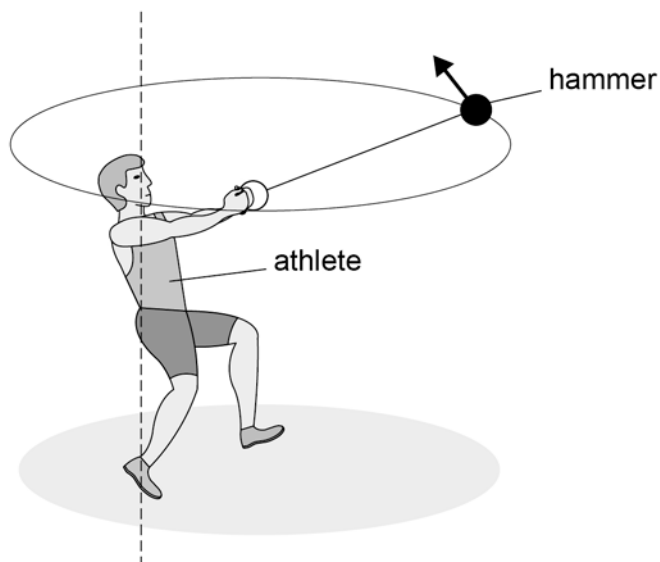
☐

0 4

In athletics, a hammer is a steel ball on a light chain. The athlete spins about a vertical axis through his body with increasing angular speed before releasing the hammer.

Figure 6 shows the athlete at the instant the hammer is released.

Figure 6



0 4 . 1

The athlete starts from rest and exerts a constant torque of 55 N m for 7.1 s before releasing the hammer.

The average combined moment of inertia of the athlete and hammer is 37 kg m^2 .

Show that the athlete's angular speed about the vertical axis at the instant of release is approximately 11 rad s^{-1} .

[3 marks]



0 4 . 2 Calculate the angular momentum of the athlete and hammer at the instant of release.

State an appropriate unit for your answer.

[2 marks]

angular momentum = _____

unit = _____

0 4 . 3 From the instant the hammer is released, the athlete no longer exerts any torque.
Assume that there is no other external torque.

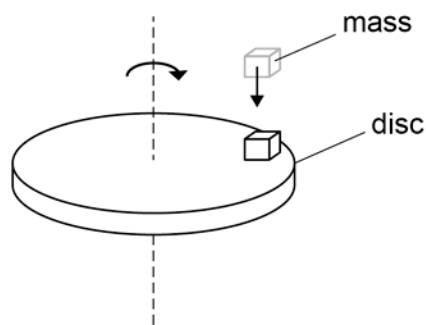
State and explain what happens to the angular speed of the athlete at the instant the hammer is released.

[3 marks]



Do not write

A disc rotates at a constant speed with no external torque acting.
A mass falls onto the rotating disc and sticks to it.



What is conserved when the mass falls onto the disc?

[1 mark]

A rotational kinetic energy

☐

B angular momentum

☐

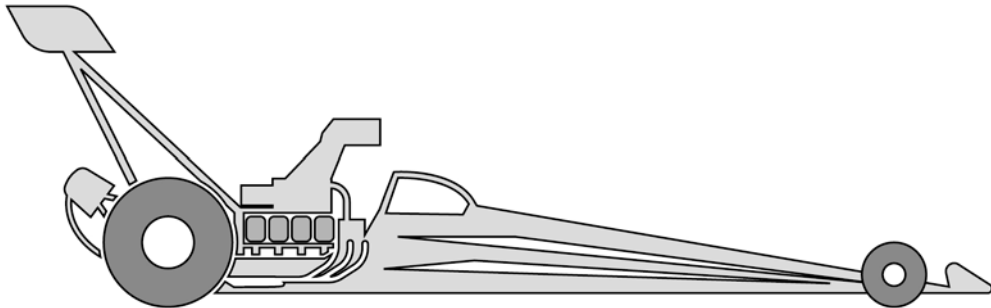
C both rotational kinetic energy and angular momentum

☐

D neither rotational kinetic energy nor angular momentum

☐

Questions **32** and **33** relate to the racing car shown in the diagram.
The front wheels each have a radius of r and a mass of m .
The rear wheels each have a radius of $2r$ and a mass of $8m$.
The wheels have similar mass distributions.



3**2**

When the car is moving without the wheels slipping, the angular velocity of the front wheels is ω .

What is the angular velocity of the rear wheels?

[1 mark]

A $\frac{\omega}{4}$



B $\frac{\omega}{2}$



C 2ω



D 4ω



3**3**

The moment of inertia of a front wheel is I .

What is the moment of inertia of a rear wheel?

[1 mark]

A $4I$



B $16I$



C $32I$



D $64I$



Turn over ►



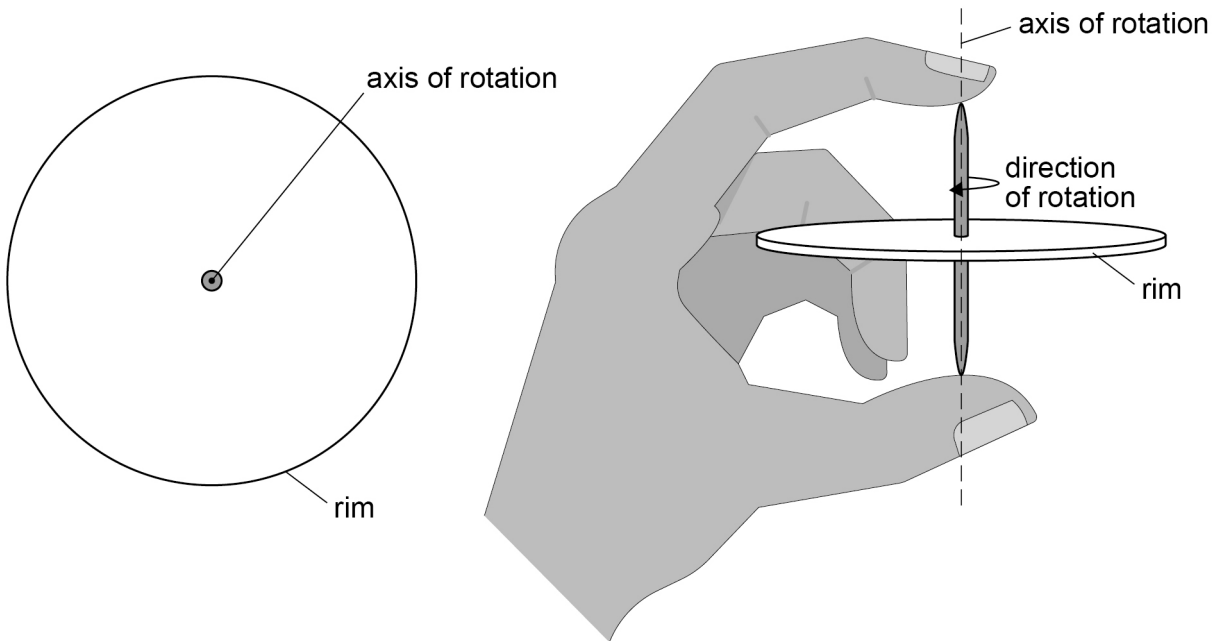
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0 2

Figure 1 shows a spinning toy rotating about its axis.

Figure 1



The moment of inertia of the toy about its axis of rotation is $3.5 \times 10^{-5} \text{ kg m}^2$.

0 2

. 1

The toy is rotating with an initial angular speed of 19.4 rad s^{-1} .
A constant frictional torque of $2.7 \times 10^{-4} \text{ N m}$ acts on the toy.
The toy decelerates to rest.

Calculate the time taken for the toy to decelerate to rest.

[2 marks]

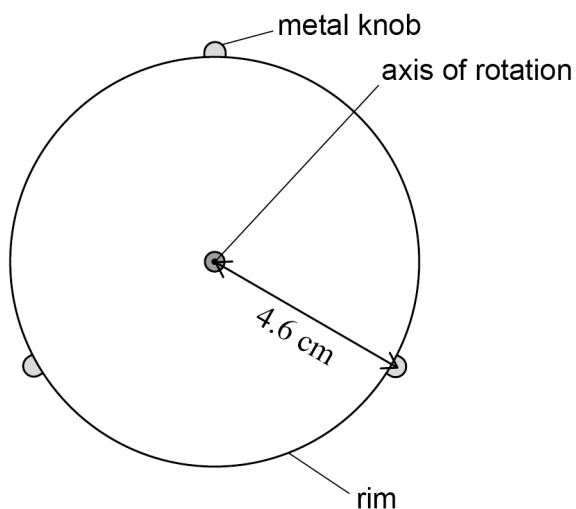
time taken = _____ s



Three identical metal knobs are added to the rim of the toy as shown in **Figure 2**.

The centre of mass of each knob is 4.6 cm from the axis of rotation.

Figure 2



The addition of the three knobs increases the moment of inertia of the toy to $8.1 \times 10^{-5} \text{ kg m}^2$ about its axis of rotation.

0 2 . 2

Calculate the mass of one knob.

[3 marks]

mass of one knob = _____ kg

Question 2 continues on the next page

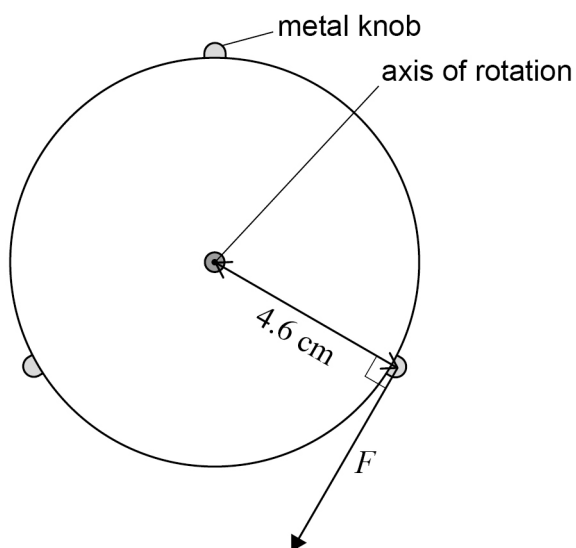
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0 2 . 3

Figure 3 shows a force F of constant magnitude acting on one of the knobs to make the toy rotate again.

Figure 3



F acts for a time of 0.12 s in a direction tangential to the rim.

The toy gains an angular momentum of $2.1 \times 10^{-4} \text{ kg m}^2 \text{ s}^{-1}$.

The frictional torque is still $2.7 \times 10^{-4} \text{ N m}$.

Calculate the magnitude of F .

[4 marks]

magnitude of $F =$ _____ N

Turn over for the next question

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outside the
box*

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ANSWER IN THE SPACES PROVIDED**

Turn over ►



Do not write Do not write

Figure 3 shows a wheel **X** held at rest on a ramp. A white sticker **R** is placed on the side of the wheel where the rim of the wheel touches the ramp. The wheel is released at time $t = 0$ and accelerates uniformly down the ramp without slipping.

Figure 3

wheel **X** at release position

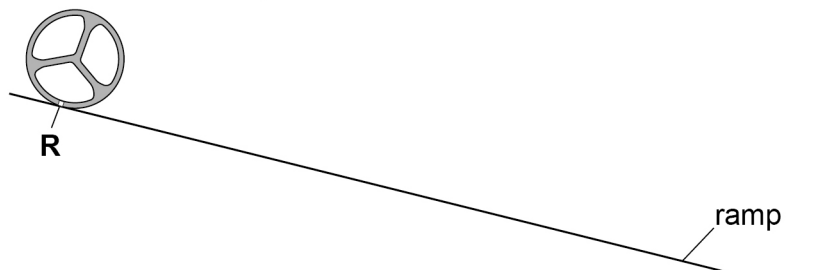
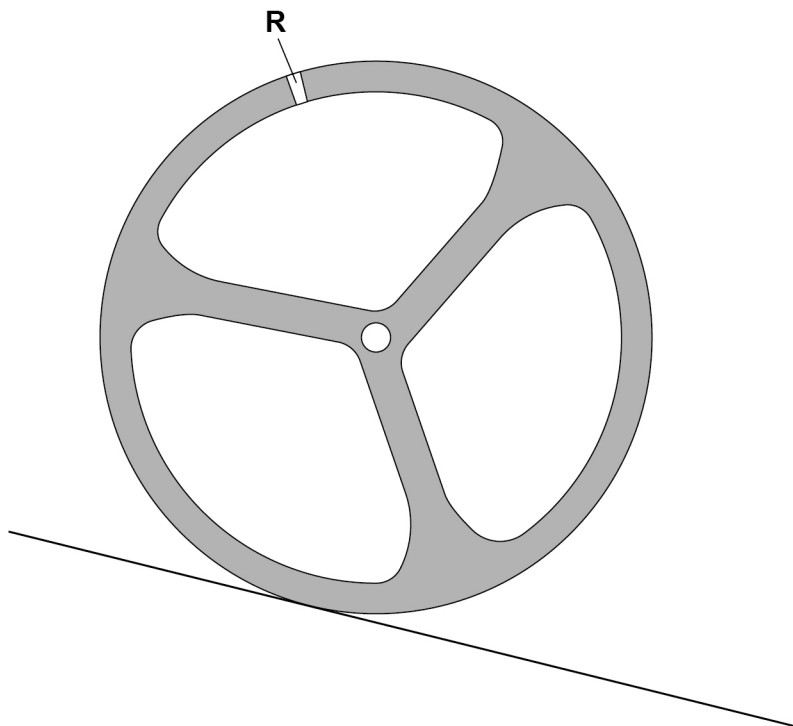


Figure 4 shows **X** at time t_1 when it has completed less than one rotation.

Figure 4



The angular acceleration of **X** as it travels down the ramp is 3.6 rad s^{-2} .

0 2 . 1

Show, using **Figure 4**, that **X** has rotated by approximately 2.6 rad at t_1 .

[2 marks]

0 2 . 2

Determine t_1 .

[1 mark]

$t_1 =$ _____ s

0 2 . 3

Calculate the angular speed of **X** as it completes its first rotation.

[2 marks]

angular speed = _____ rad s^{-1}

Question 2 continues on the next page

Turn over ►

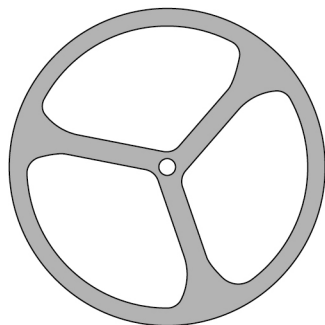


0 2 . 4

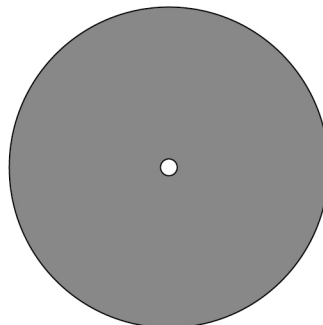
Figure 5 shows wheel **X** and a uniform disc **Y**.
Y has the same mass and the same diameter as **X**.

Figure 5

wheel **X**



uniform disc **Y**



X and **Y** are released from rest and allowed to accelerate down the same ramp.

Compare the angular acceleration of **X** with the angular acceleration of **Y**.

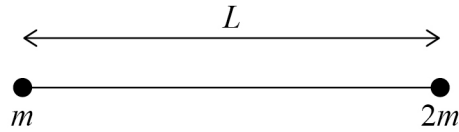
[3 marks]



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A system consists of two point masses m and $2m$ connected by a rigid rod of negligible mass.

The distance between the point masses is L .



What is the moment of inertia about the centre of mass of the system?

[1 mark]

A $\frac{2mL^2}{3}$

☐

B $\frac{3mL^2}{4}$

☐

C mL^2

☐

D $\frac{4mL^2}{3}$

☐

Turn over for the next question



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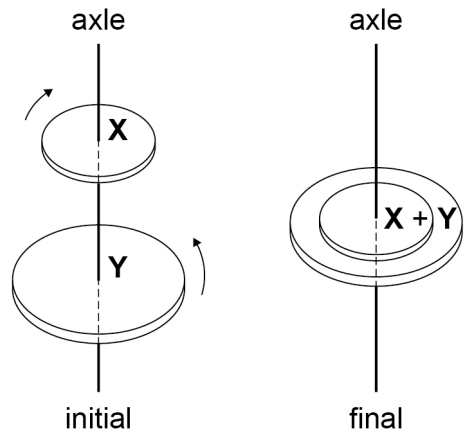
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1

Two discs **X** and **Y** rotate freely on the same axle at different angular speeds. Initially, **X** rotates clockwise at 48 rad s^{-1} and **Y** rotates anticlockwise at 22 rad s^{-1} .

X falls onto **Y** and the two discs rotate together with the same angular speed.

The moment of inertia of **X** is 1.2 kg m^2 .
The moment of inertia of **Y** is 1.6 kg m^2 .



What is the final angular speed and direction of rotation of the two discs?

[1 mark]

	Final angular speed / rad s^{-1}	Direction of rotation
A	8	clockwise
B	22	clockwise
C	33	anticlockwise
D	93	anticlockwise

☐☐☐☐

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2 3 1 X P H 0 4

Figure 10 shows a system made of two metal spheres attached by rigid rods to a frictionless hinge. The hinge is at the top of a vertical pole.

Figure 10

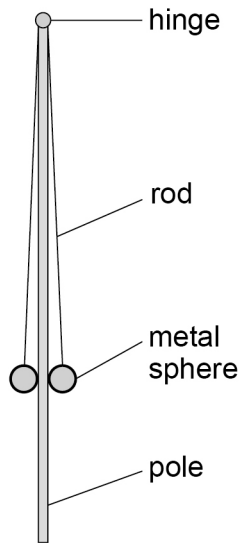


Figure 11

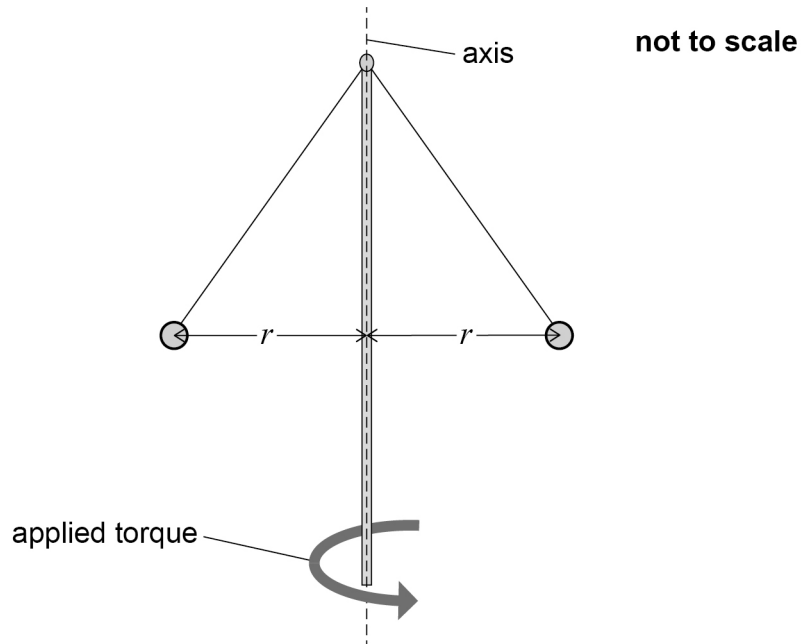


Figure 11 shows a torque being applied to the pole, making it rotate about its axis.

The torque is removed when the distance between the centre of mass of each sphere and the axis is r .

The mass of each sphere is 0.24 kg. For the position shown in **Figure 11**, the rotating system has a moment of inertia of $2.2 \times 10^{-3} \text{ kg m}^2$ and rotational kinetic energy of 59 mJ about the axis.

Assume that the spheres behave as point masses. The pole, rods and hinge each have negligible mass.

Calculate r .

[2 marks]

$r =$ _____ m



0 6 . 2 Calculate the angular velocity of the spheres.

[2 marks]

angular velocity = _____ rad s⁻¹

A constant frictional torque is now applied to the rotating pole. When the system has come to rest the arrangement returns to that shown in **Figure 10**.

0 6 . 3 Explain why the angular deceleration is **not** constant as the spheres come to rest.

[2 marks]

0 6 . 4 Explain why the work done by the frictional torque is greater than 59 mJ.

[2 marks]



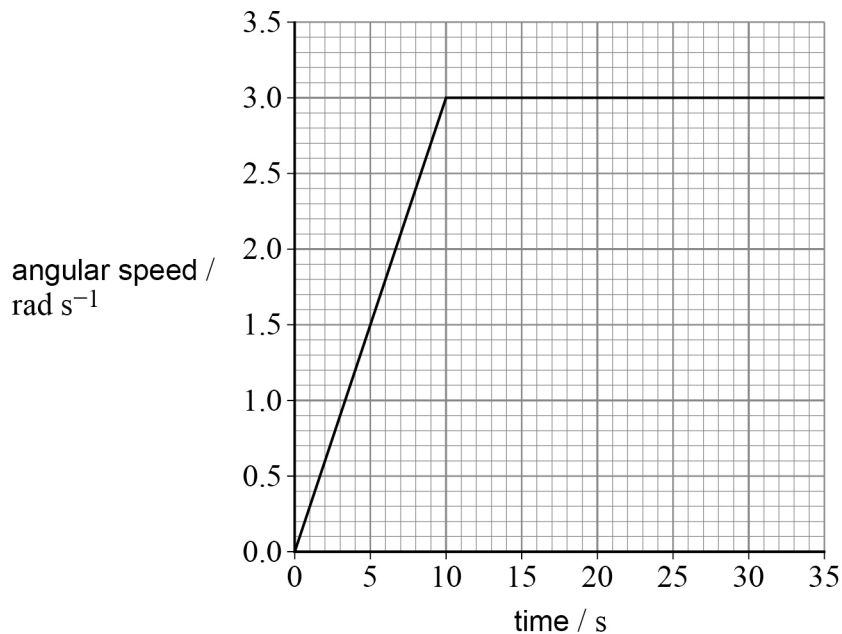
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A flywheel is accelerated from rest.

The angular speed of the flywheel varies with time as shown.



What is the average angular speed of the flywheel over the 35 s?

[1 mark]

A 1.5 rad s^{-1}

☐

B 2.0 rad s^{-1}

☐

C 2.3 rad s^{-1}

☐

D 2.6 rad s^{-1}

☐

2**2**

A flywheel is initially rotating at 5.0 rad s^{-1} . It comes to rest after a constant angular deceleration of 0.25 rad s^{-2} .

How many revolutions does the flywheel make as it decelerates to rest?

[1 mark]

A 8

☐

B 16

☐

C 20

☐

D 50

☐

[illegible]

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0 5

A wind turbine has a rotor with a diameter of 135 m.

0 5 . 1

Calculate, in MW, the maximum power available from the air to the turbine when the wind speed is 11.4 m s^{-1} .

density of air = 1.2 kg m^{-3}

[2 marks]

maximum power = _____ MW

0 5 . 2

State **two** reasons why the total kinetic energy of the wind passing through the turbine is **not** available to the turbine.

[2 marks]

1 _____

2 _____

The rotor accelerates from rest to its maximum angular speed of 1.41 rad s^{-1} .

0 5 . 3

Calculate, in revolutions per minute, the maximum frequency of rotation of the rotor.

[1 mark]

frequency = _____ revolutions per minute

Turn over ►



The rotor is connected to an electric generator.
The wind exerts a constant torque on the rotor.

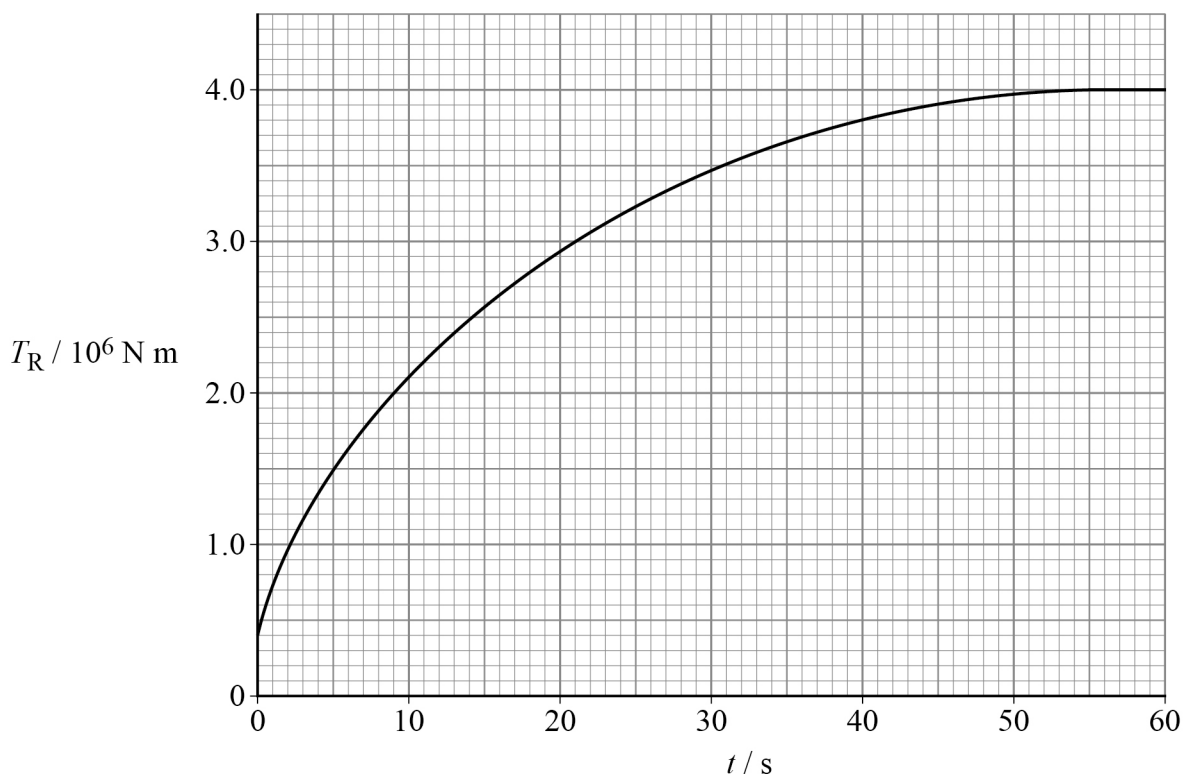
Two types of resistive torque are exerted on the rotor:

- a constant frictional resistive torque of $4.0 \times 10^5 \text{ N m}$
- a varying resistive torque exerted by the generator.

The total resistive torque is T_R .

Figure 3 shows the variation of T_R with time t while the rotor is accelerating. The rotor achieves its maximum angular speed of 1.41 rad s^{-1} at $t = 55 \text{ s}$.

Figure 3



0 5 . 4

Calculate, in MJ, the energy that the wind transfers to the turbine between $t = 55 \text{ s}$ and $t = 60 \text{ s}$.

Use data from **Figure 3**.

[2 marks]

energy = _____ MJ



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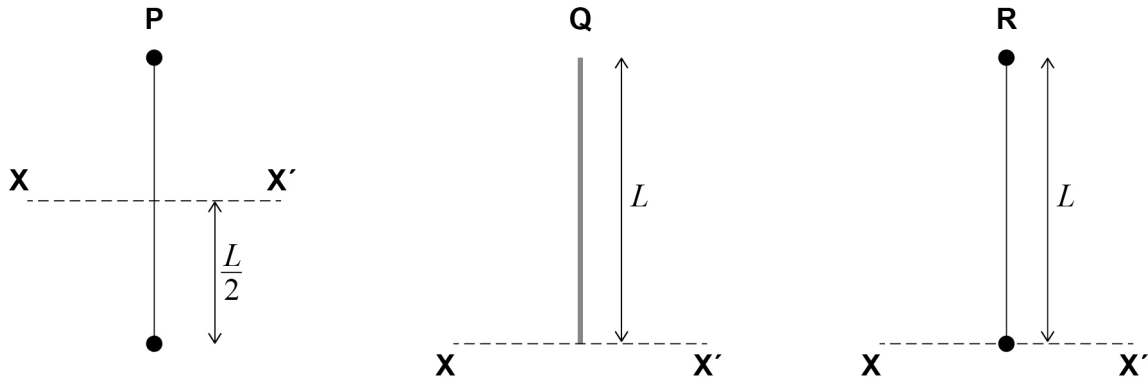
1 8

P, **Q** and **R** are objects with the same mass m and the same length L .

P and **R** each consist of a light rod with a point mass of mass $\frac{m}{2}$ at each end.

Q is a uniform rod of mass m . **Q** has a moment of inertia of $\frac{1}{3}mL^2$.

X-X' is the axis of rotation for each object.



Which shows the objects in order of increasing moment of inertia?

[1 mark]

A P, Q, R ☐

B P, R, Q ☐

C R, P, Q ☐

D R, Q, P ☐

Turn over for the next question

Turn over ►



Don't write

A wheel has an initial angular velocity of 10 rad s^{-1} .

The wheel experiences an angular acceleration of -2.0 rad s^{-2} until its angular velocity is -20 rad s^{-1} .

How many revolutions does the wheel make during this change of angular velocity?

[1 mark]

A 12

☐

B 20

☐

C 75

☐

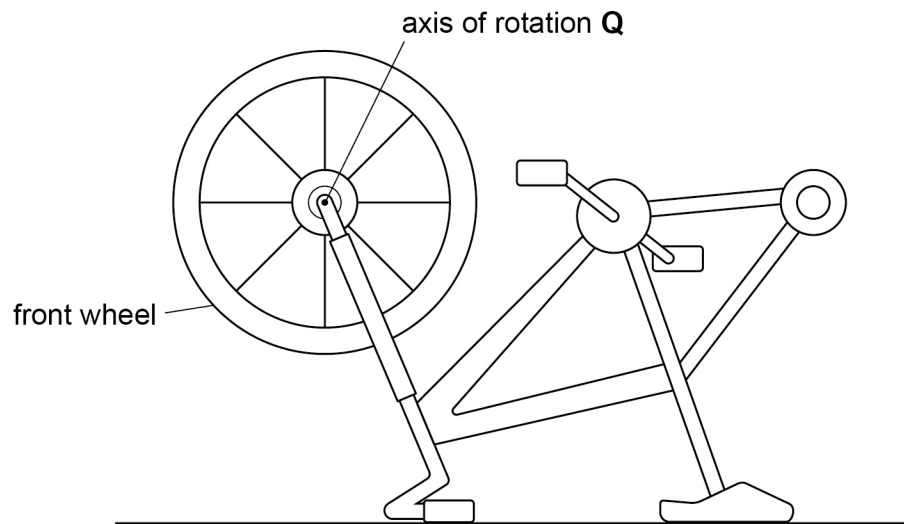
D 125

☐

A student investigates the frictional torque on the wheel of a bicycle.

Figure 5 shows a bicycle that has been turned upside down and placed on level ground. The back wheel is removed. The front wheel is not connected to the pedals. The axis of rotation **Q** of the front wheel is horizontal.

Figure 5



05.1

The wheel has an initial angular speed of 21 rad s^{-1} .

A frictional torque of $3.6 \times 10^{-3} \text{ N m}$ acts on the wheel.
The wheel takes 481 revolutions to come to rest.

Calculate the moment of inertia of the wheel.

[3 marks]

moment of inertia = _____ kg m^2

Question 5 continues on the next page

Turn over ►



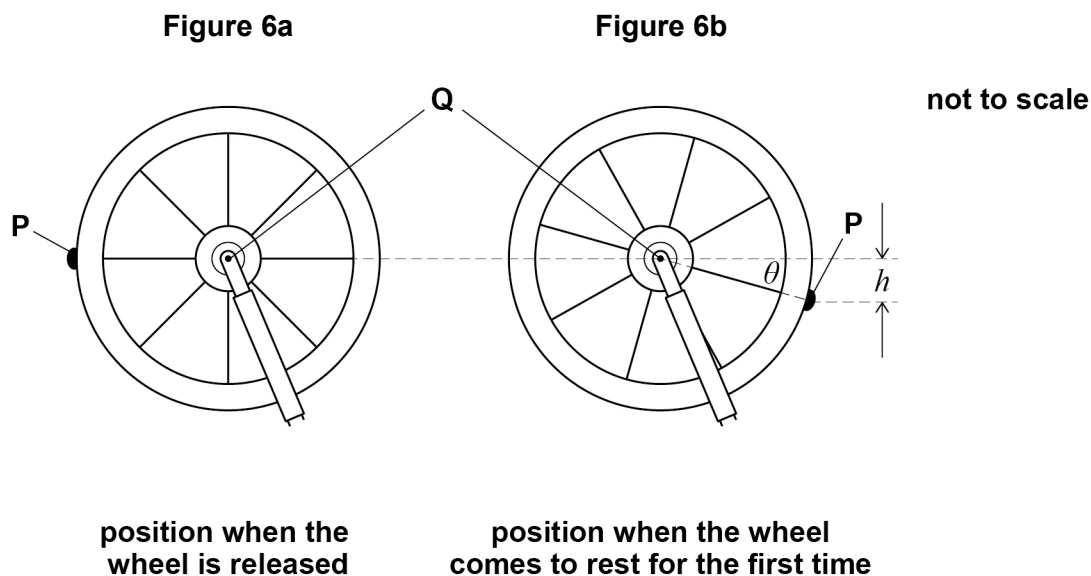
The student then reduces the frictional torque on the wheel and attaches a mass **P** to the outside of the wheel.

The mass of **P** is 0.017 kg.

The edge of the wheel is held with **P** at the same height above the ground as **Q**, as shown in **Figure 6a**. The edge is then released and the wheel begins to rotate.

Figure 6b shows the wheel when it comes to rest for the first time. The angle θ between the horizontal and the line from **Q** through the centre of mass of **P** is measured.

The vertical distance h from the centre of mass of **P** to its starting position is also measured.



The work done by the frictional torque is equal to the change in the gravitational potential energy of **P**.

0 5 . 2

The student obtains the following readings:

$$\theta = 5^\circ$$

$$h = 4.8 \text{ cm}$$

Calculate the new frictional torque.

[4 marks]

new frictional torque = N m

0	5	.	3
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The student now repeats the investigation.

Describe how repeated measurements can be used to determine the absolute uncertainty in the value of the frictional torque.

Assume that the uncertainty in the mass of **P** is negligible.

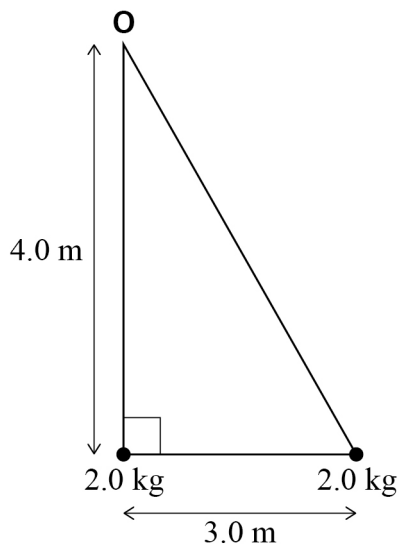
[4 marks]

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The diagram shows a light triangular card with point masses of 2.0 kg attached at two corners.

not to scale



The system is completed by adding a third point mass of 2.0 kg at a distance of 1.0 m from O .

What is the moment of inertia of the system about O ?

[1 mark]

A 16 kg m^2 ☐

B 20 kg m^2 ☐

C 82 kg m^2 ☐

D 84 kg m^2 ☐



Do not write

A disc is rotating.

At time $t = 0$ the disc has an angular displacement of zero and an angular velocity of 8.0 rad s^{-1} anticlockwise.

The disc accelerates uniformly to an angular velocity of 12.0 rad s^{-1} clockwise. As a result of the acceleration, the final angular displacement of the disc is 80 rad clockwise.

What is the angular acceleration?

[1 mark]

A 0.50 rad s^{-2} clockwise

☐

B 0.50 rad s^{-2} anticlockwise

☐

C 0.90 rad s^{-2} clockwise

☐

D 0.90 rad s^{-2} anticlockwise

☐

[illegible]

[illegible]

[illegible]

[illegible]

0 3

Table 1 gives the nuclear radius R for five different nuclides.

Table 1

Nuclide	R / fm		
${}^4_2\text{He}$	1.68		
${}^{15}_7\text{N}$	2.61		
${}^{23}_{11}\text{Na}$	2.99		
${}^{32}_{18}\text{Ar}$	3.35		
${}^{72}_{36}\text{Kr}$	4.41		

Theory suggests that R is related to the nucleon number A by

$$R = R_0 A^{\frac{1}{3}}$$

where R_0 is a constant.

0 3 . 1

Show that the data in **Table 1** are consistent with this equation.

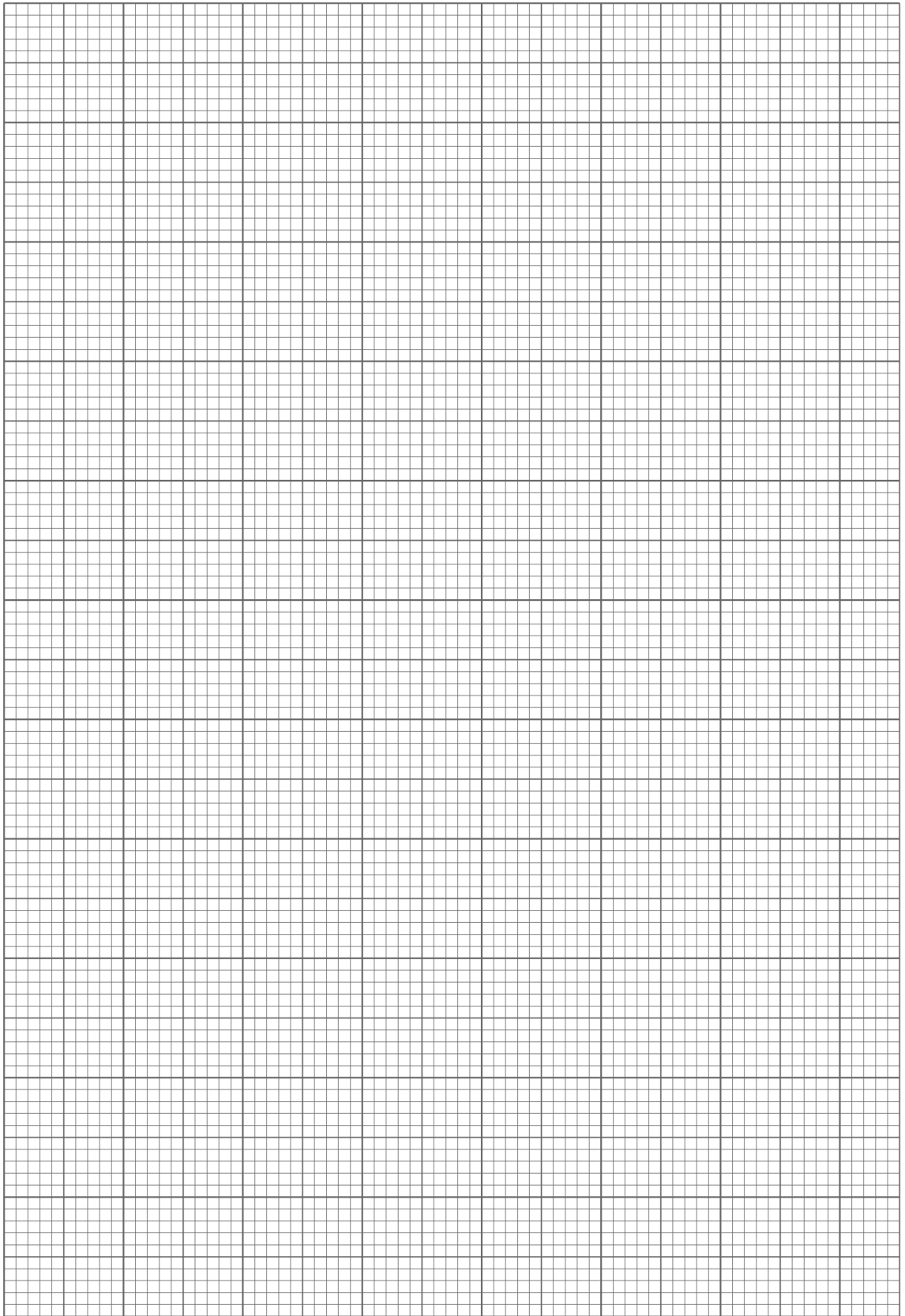
Plot a straight-line graph on **Figure 3** as part of your answer. Use one or more of the blank columns in **Table 1** for any data that you wish to calculate.

Explain how your graph is consistent with the equation.

[5 marks]



Figure 3



Question 3 continues on the next page

Turn over ►



0 3 . 2

The data in **Table 1** were obtained by electron diffraction.
Using the same method, the radius of a palladium nucleus was found to be 5.03 fm.

Determine, using your graph in **Figure 3**, the number of nucleons in this palladium nucleus.

[2 marks]

number of nucleons = _____

0 3 . 3

The Coulomb equation is used to estimate the closest approach of alpha particles to a sodium ${}_{11}^{23}\text{Na}$ nucleus.

Calculate the maximum kinetic energy of an alpha particle that has the closest approach of 25 fm to the sodium nucleus.
Assume that the sodium nucleus does not recoil.

[3 marks]

maximum kinetic energy = _____ J

0 3 . 4

Explain why the radius of the sodium nucleus obtained by electron diffraction is different from the estimate obtained by the closest approach of alpha particles.

[1 mark]

Turn over for the next question

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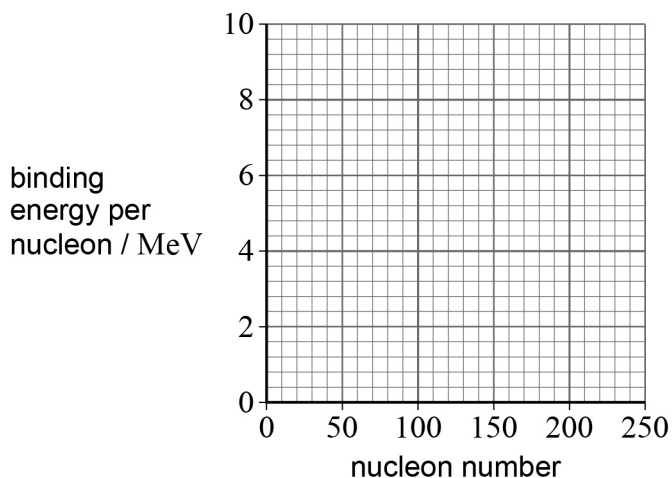
0 4 . 1 State what is meant by the binding energy of a nucleus.

[1 mark]

0 4 . 2 Sketch, on the axes in **Figure 4**, a plot of binding energy per nucleon against nucleon number.

[2 marks]

Figure 4



0 4 . 3 A large nucleus **X** undergoes a fission reaction to form two smaller nuclei **Y** and **Z**.

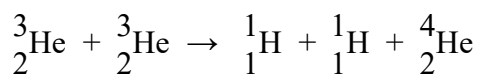
State how the binding energy per nucleon of **X** compares with the binding energy per nucleon of **Y** and of **Z**.

Explain your answer.

[2 marks]



A fusion reaction that occurs in a star is



The binding energy per nucleon of ${}^3_2\text{He} = 2.57 \text{ MeV}$.

The binding energy per nucleon of ${}^4_2\text{He} = 7.07 \text{ MeV}$.

0 4 . 4

Explain why the binding energy per nucleon of ${}^1_1\text{H}$ is zero.

[1 mark]

0 4 . 5

The reaction occurs 9.2×10^{37} times per second in the star.

Calculate the total power output from these reactions within the star.

[4 marks]

total power output = _____ W



0 5

A potter is a person who forms pots from a material called clay. The pots can then be hardened for use in cooking.

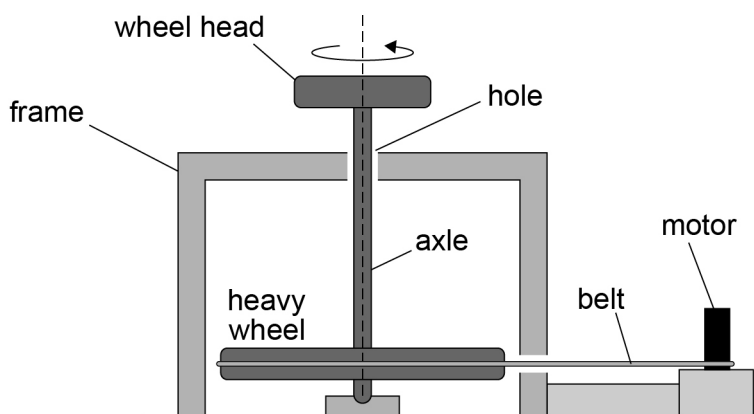
Figure 5 shows a rotating system used by the potter. The system is inside a stationary frame.

The system is made up of three parts that are connected together:

- a horizontal wheel head
- a vertical axle
- a horizontal heavy wheel.

The horizontal heavy wheel is attached to the wheel head by the vertical axle. The axle passes through a hole in the frame.

Figure 5



A motor is used to turn the system. The motor is connected to the heavy wheel by a belt.

The moment of inertia of the system about the axle is 35.0 kg m^2 .

0 5 . 1

At one instant, the force applied by the belt makes the system rotate at a constant angular speed. The rotational kinetic energy of the system is 933 J .

Calculate the time taken for the system to complete one revolution at this angular speed.

[2 marks]

time = _____ s



0 5 . 2

The potter uses the motor to increase the speed of the system.

The belt applies a tangential force F to the heavy wheel at a distance of 0.68 m from the centre of the axle.

As F is being applied, the system moves through an angle of 17° and increases its rotational kinetic energy from 933 J to 947 J.

A constant frictional torque of 18 N m acts on the system.

Calculate F .

[5 marks]

$F =$ _____ N

Question 5 continues on the next page



The motor accelerates the system to a constant angular speed of 7.6 rad s^{-1} .
The belt is then detached from the heavy wheel and the angular speed does not change.

The potter then drops a piece of clay vertically onto the centre of the wheel head.
The clay sticks to the centre of the wheel head.
The total moment of inertia of the system and the clay about the axle is 39.7 kg m^2 .

0 5 . 3

Calculate the angular speed of the system immediately after the clay sticks to the wheel head.
Assume that there is no net external torque on the clay and system.

[2 marks]

angular speed = _____ rad s^{-1}



0 5 . 4

The clay and system start to slow down due to the frictional torque.

At a time $t = 0$ the clay and system are rotating at 5.0 rad s^{-1} .

At time $t = t_1$ the clay and system come to rest. Between $t = 0$ and $t = t_1$ the system rotates through an angle θ .

The frictional torque on the system is 18 N m as it slows down.

Calculate

- t_1
- θ

[3 marks]

$t_1 =$ _____ s
 $\theta =$ _____ rad

12

Turn over for the next question

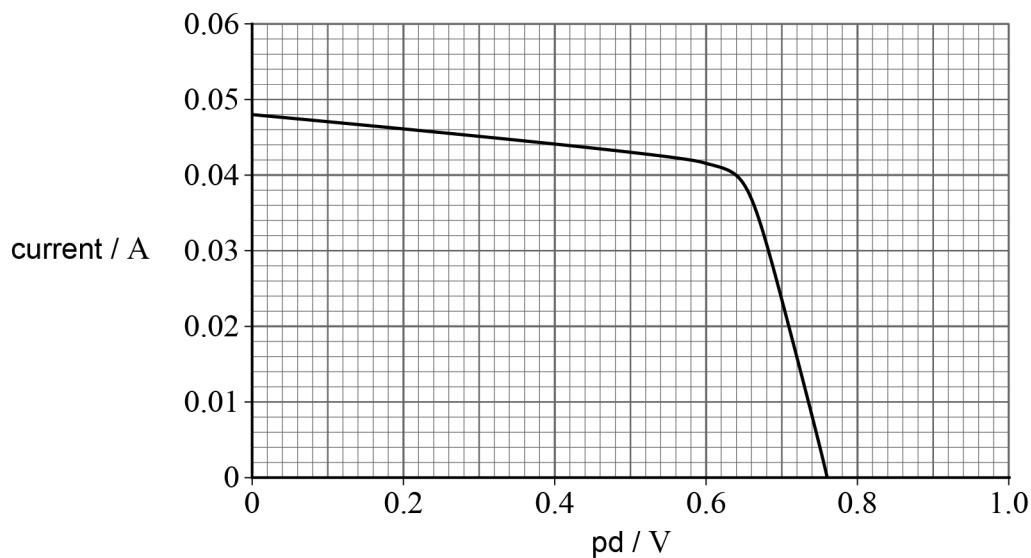
Turn over ►



0 6

Figure 6 shows the variation of current in a solar cell with potential difference (pd) across the cell.

Figure 6



0 6 . 1

Deduce the current and pd when the power output of the cell is at its maximum value.

[1 mark]

current = _____ A

pd = _____ V

The cell is one of many identical cells that form a solar array on a spacecraft that orbits the Sun.

The intensity of the Sun's power at the orbit of the spacecraft is 720 W m^{-2} .

0 6 . 2

Calculate the distance between the centre of the Sun and the spacecraft.

power output of Sun = $3.8 \times 10^{26} \text{ W}$

[2 marks]

distance = _____ m



0 6 . 3

The spacecraft requires a power of 650 W.

To prevent the spacecraft overheating, 75% of the energy from the Sun that is incident on the solar array is reflected back into space and not transferred to the solar array.

45% of the energy that is transferred to the solar array becomes electrical energy.

Each cell has an area of 20 cm^2 .

Determine the number of solar cells needed to power the spacecraft.

[4 marks]

number of cells = _____

7

Turn over for the next question

Turn over ►

Section B

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.



If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional pages for this working.

0 8

A sample of gas is at a constant temperature.

The gas contains two isotopes **X** and **Y** of the same element.

X has fewer neutrons than **Y**.

Which statement is correct?

[1 mark]

A The rms speed of the molecules of **X** is greater than the rms speed of the molecules of **Y**.

☐

B The rms speed of the molecules of **Y** is greater than the rms speed of the molecules of **X**.

☐

C Molecules of **X** have a greater mean kinetic energy than molecules of **Y**.

☐

D Molecules of **Y** have a greater mean kinetic energy than molecules of **X**.

☐

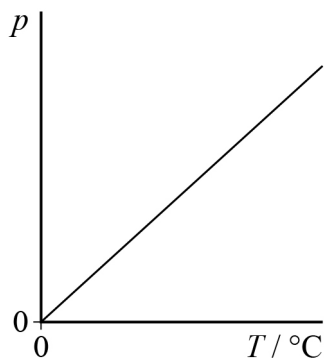
0 9

The volume of a fixed mass of an ideal gas remains constant.
The temperature of the gas is increased.

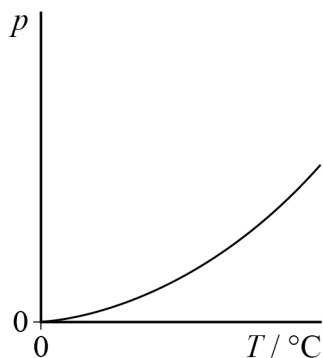
Which graph shows the variation of the pressure p with temperature T in $^{\circ}\text{C}$?

[1 mark]

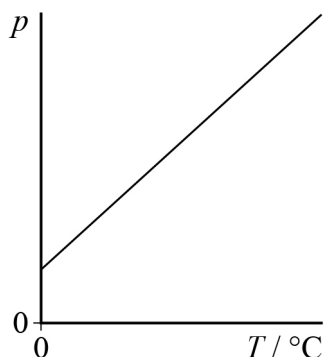
A



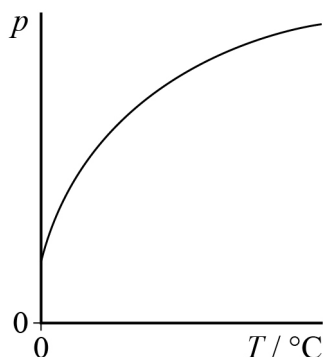
B



C



D



A

☐

B

☐

C

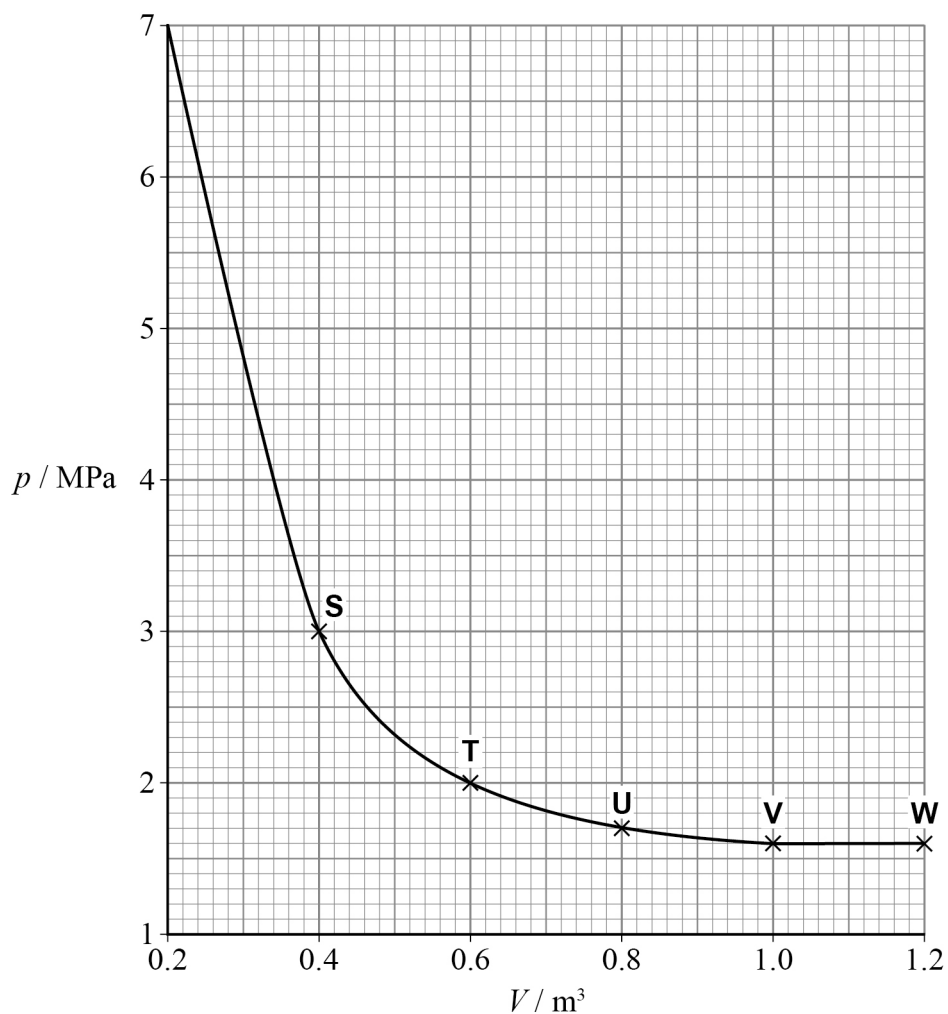
☐

D

☐


1 0 The graph shows the variation of pressure p with volume V for a fixed mass of an ideal gas.

S, T, U, V and **W** are points on the graph.



For which pair of points is the temperature of the gas the same?

[1 mark]

- A** **S** and **T** ☐
- B** **T** and **U** ☐
- C** **U** and **V** ☐
- D** **V** and **W** ☐

- 1 1** Four gas molecules are moving along the same straight line with velocities of $+250 \text{ m s}^{-1}$, -25 m s^{-1} , -100 m s^{-1} and -125 m s^{-1} .

What is the rms speed of the molecules?

[1 mark]

A 0 ☐

B 74 m s^{-1} ☐

C 125 m s^{-1} ☐

D 149 m s^{-1} ☐

- 1 2** A system contains a fixed mass of an ideal gas.

Which combination of processes **always** produces a decrease in the internal energy of the system?

[1 mark]

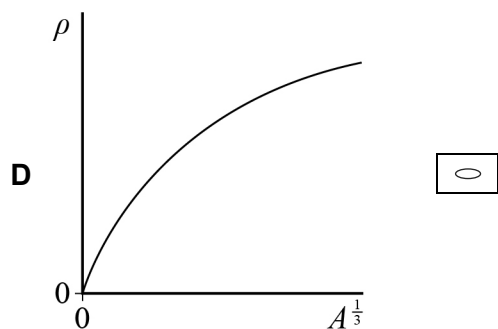
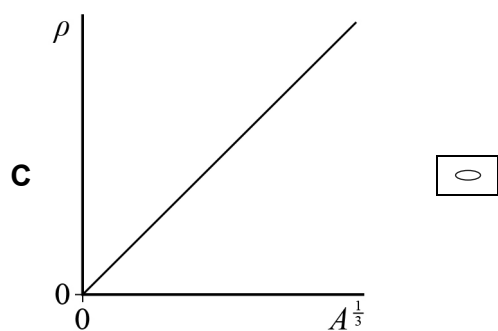
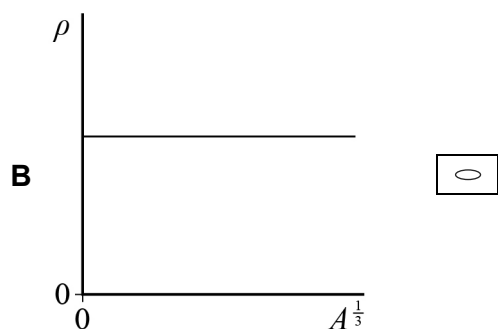
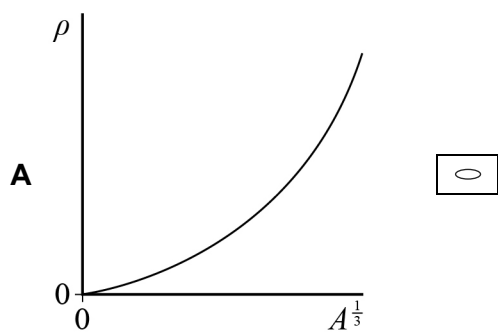
	Work is done...	Energy is transferred by heating...	
A	by the system.	to the system.	☐
B	by the system.	from the system.	☐
C	on the system.	to the system.	☐
D	on the system.	from the system.	☐

1 3

Nuclear density is ρ .
Nucleon number is A .

Which graph shows the variation of ρ with $A^{\frac{1}{3}}$?

[1 mark]



Questions **14** and **15** are about four samples of different materials.

The table shows the properties of samples **A**, **B**, **C** and **D**.

Sample	U-value / $\text{W m}^{-2} \text{K}^{-1}$	Thickness / mm	Area / m^2
A	1.0	30	0.04
B	1.0	20	0.05
C	2.0	30	0.04
D	2.0	20	0.05

The temperature difference across each sample is the same.

1 4

Energy is transferred by conduction through each sample.

Which sample has the greatest rate of energy transfer by conduction?

[1 mark]

- A** ☐
- B** ☐
- C** ☐
- D** ☐

1 5

Which sample has the greatest thermal conductivity?

[1 mark]

- A** ☐
- B** ☐
- C** ☐
- D** ☐



1 6 Two discs **X** and **Y** are rotating about an axis.

The table gives information about the mass and diameter of **X** and **Y**.

	Mass of disc	Diameter of disc
X	m	d
Y	$2m$	$0.5d$

For a disc of mass M and radius r , the moment of inertia about this axis is $\frac{1}{2}Mr^2$.

X and **Y** are rotating with the same angular momentum.

The rotational kinetic energy of **X** is E .

What is the rotational kinetic energy of **Y**?

[1 mark]

A $\frac{E}{2}$ ☐

B E ☐

C $2E$ ☐

D $4E$ ☐

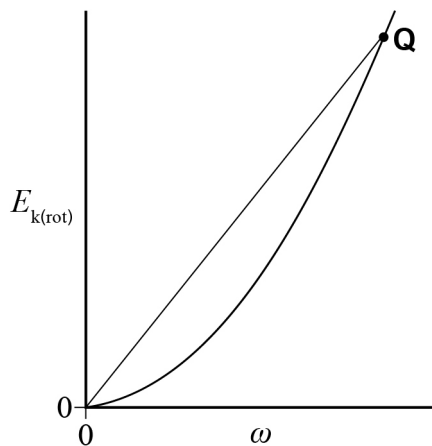
Turn over for the next question



1 7

The graph shows the variation in rotational kinetic energy $E_{\text{k(rot)}}$ with angular speed ω for a rotating object.

The moment of inertia of the object is I and its angular momentum is P_{A} .



A straight line is drawn from the origin to point Q on the graph.

What does the gradient of this line represent?

[1 mark]

A $\frac{1}{2} P_{\text{A}}$ ☐

B P_{A} ☐

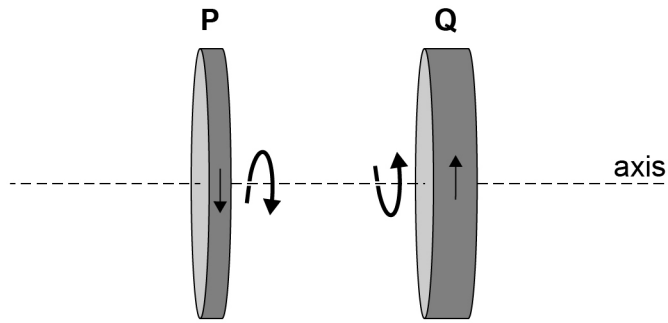
C $\frac{1}{2} I$ ☐

D I ☐



1 8

The diagram shows two discs **P** and **Q** rotating freely about a common axis.



P has a moment of inertia about its axis of 12 kg m^2 and an angular velocity of 8.0 rad s^{-1} clockwise.

Q has a moment of inertia about its axis of 15 kg m^2 and an angular velocity of 6.0 rad s^{-1} anticlockwise.

P and **Q** come into contact. They reach a new constant angular velocity. They stick together and do not slip.

Which row shows the new constant angular velocity?

[1 mark]

	Magnitude / rad s^{-1}	Direction	
A	0.22	clockwise	☐
B	0.22	anticlockwise	☐
C	6.9	clockwise	☐
D	6.9	anticlockwise	☐

Turn over for the next question

Turn over ►



1 9**M** is a moderator in a thermal nuclear reactor.Which row shows the ideal properties of a nucleus of **M**?**[1 mark]**

	Probability of absorbing a neutron	Mass compared to mass of a neutron	
A	low	much greater	☐
B	low	similar	☐
C	high	much greater	☐
D	high	similar	☐

2 0

Fuel rods are removed from a nuclear fission reactor and stored in water for several months.

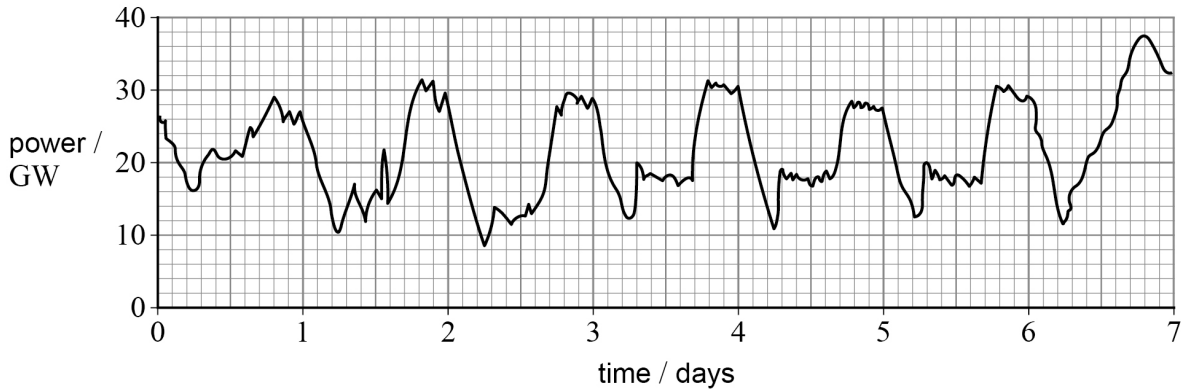
This is because the water:

[1 mark]**A** stops the remaining fuel from undergoing fission in the fuel rods.☐**B** cools the fuel rods that still generate heat.☐**C** combines with the elements in the fuel and reduces their radioactivity.☐**D** dissolves the fuel which makes it easier to transport for reprocessing.☐

Questions **21** and **22** refer to the graph below.

A country's electricity is generated partly by base-power stations and partly by back-up power stations.

The graph shows the total power provided by the power stations over a period of seven days.



2 1 Which statement is consistent with the graph?

[1 mark]

- A** Back-up power stations always provide less than half of the maximum output power.
- B** The base-power stations provide more than 20 GW of power.
- C** The daily variation is due to the supply from solar power.
- D** The back-up power stations can be brought on line quickly.

☐
☐
☐
☐

2 2 What is an order of magnitude estimate for the energy supplied during the seven-day period?

[1 mark]

A 10^{11} J

☐

B 10^{13} J

☐

C 10^{16} J

☐

D 10^{19} J

☐



[illegible]

[illegible]